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Bugs Found

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Bugs Fixed

This report documents the comprehensive system audit conducted on the FieldComms Incident Management System v1.0 prior to beta deployment on Pi 5 hardware. The audit covered all 30 HTML pages, 12 Python services, 14 systemd units, the nginx configuration, the install and update scripts, and all configuration files. **All 9 bugs found during the audit have been corrected. Zero open issues remain.**

SECTION 1 — CRITICAL BUGS (All Fixed)

Critical bugs are defects that would prevent features from working when accessed from any device other than the Pi itself, or would cause data to be silently lost. All critical bugs have been corrected.

#1 localhost in API URLs — 19 pages affected

CRITICAL

FIXED

Area All HTML pages / API calls

Issue Found

19 HTML pages used `http://localhost:5050/5051/5055/5056` as the API base URL. On the Pi itself (localhost = 127.0.0.1) this works, but every other device on EMCOMM-NET resolves localhost to itself, not the Pi. All API calls from tablets, phones, and laptops would return connection refused errors.

Resolution

All 19 pages updated to use `http://192.168.50.1:PORT`. Note: `tactical.html` intentionally retains `localhost:8080/8082` for Graywolf/YAAC since those APRS services are Pi-local.

#2 Dead Man's Switch — missing /api/ prefix on all endpoints

CRITICAL

FIXED

Area `html/deadmans.html`

Issue Found

`deadmans.html` called `/dms/state`, `/dms/arm`, `/dms/reset`, `/dms/disarm`. The `fcc_lookup_server` expects `/api/dms`, `/api/dms/arm`, `/api/dms/reset`, `/api/dms/disarm`. The DMS would load blank, and all `arm/reset/disarm` actions would return 404 silently.

Resolution

All four DMS fetch calls corrected to use the `/api/` prefix.

#3 Resource Map — missing /api/ prefix

CRITICAL

FIXED

Area `html/resmap.html`

Issue Found

`resmap.html` called `${API}/resmap` for both GET and PUT. Server expects `/api/resmap`. The resource tracking map would load blank and all unit position saves would fail silently.

Resolution

Both fetch calls corrected to `${API}/api/resmap`.

#4 ICS form POST paths wrong — `nts.html`, `ics213.html`, `ics214.html`

CRITICAL

FIXED

Area `html/nts.html` · `html/ics213.html` · `html/ics214.html`

Issue Found

`nts.html` posted to `${API}/forms/nts`, `ics213.html` to `${API}/forms/ics213`, `ics214.html` to `${API}/forms/ics214`. The server POST handler is at `/api/forms`. All three forms were being submitted to non-existent endpoints — data was lost silently.

Resolution

All three corrected to POST to `${API}/api/forms`.

#5 Dashboard mode sections hardcoded visible on load

CRITICAL

FIXED

Area html/index.html

Issue
Found

Five amateur-section divs had class="mode-section amateur-section visible" hardcoded. On page load, all Amateur Radio cards displayed regardless of the user's saved mode, then flickered when setMode() ran on DOMContentLoaded.

Resolutio
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Removed hardcoded visible class. setMode() now controls visibility exclusively on DOMContentLoaded.

#6 Backup service — wrong web root

CRITICAL

FIXED

Area systemd/fieldcomms-backup@.service

Issue
Found

The USB backup service rsynced from /var/www/html/ — the default nginx web root, not the FieldComms web root. USB backups would contain an empty directory.

Resolutio
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Corrected to /opt/fieldcomms/html/.

SECTION 2 — MEDIUM BUGS (All Fixed)

Medium bugs affect system function or deployment completeness but do not cause silent data loss. All have been corrected.

#7 No firewall rules in install.sh

MEDIUM

FIXED

Area scripts/install.sh

Issue
Found

install.sh never configured ufw. On Raspberry Pi OS Lite the default firewall may block ports 5050, 5051, 5055, 5056, 8080–8090, causing all API calls to time out from EMCOMM-NET devices after deployment.

Resolutio
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Added ufw configuration step opening all 11 required ports: 22 (SSH), 80 (nginx), 5050 (FCC API), 5051 (Health), 5055 (ICS), 5056 (Refs), 8080 (Graywolf), 8081 (Kiwix), 8082 (YAAC), 8083 (Tiles), 8090 (Pat).

#8 nginx config — wrong web root and missing tile server proxy

MEDIUM

FIXED

Area udev/nginx-fieldcomms.conf

Issue
Found

nginx config pointed to /var/www/html as the web root (the Debian default). FieldComms files are installed at /opt/fieldcomms/html. Also missing: tile server proxy (port 8083), CORS OPTIONS preflight handling, health API path was /health/ (server expects /health-api/), and no access/error log directives.

Resolutio
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Web root corrected to /opt/fieldcomms/html. Tile server proxy added at /tiles/. CORS preflight handling added (OPTIONS → 204). Health API path corrected. Log directives added.

#9 hostapd/dnsmasq config — wrong network architecture

MEDIUM

FIXED

Area udev/hostapd.conf · udev/dnsmasq-fieldcomms.conf · scripts/install.sh

Issue
Found

Install script and config files configured the Pi as a Wi-Fi access point using hostapd and dnsmasq. The actual hardware uses an ASUS RT-BE58 Go router as the AP. Running hostapd would conflict with the router and break Wi-Fi on EMCOMM-NET.

Resolutio
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hostapd.conf and dnsmasq.conf archived as .old files. install.sh AP setup block replaced with nmcli static IP configuration. New NetworkManager-static-ip.conf documents the correct Pi static IP setup.

SECTION 3 — SYSTEM VERIFICATION

The following items were verified correct during the audit and require no changes.

ITEM	STATUS	NOTES
All 12 Python scripts	✓ PASS	Zero syntax errors; all compile clean

All 14 systemd service units	✓ PASS	All ExecStart scripts exist at correct paths
Database schema (20 tables)	✓ PASS	All tables referenced by servers are defined in db.py
NWS alert coordinates	✓ PASS	All 7 files use 42.3153/-88.4473 (McHenry County, IL)
Port assignments (no conflicts)	✓ PASS	5050/5051/5055/5056/8080/8081/8082/8083/8090 all unique
Inter-page navigation links	✓ PASS	All href targets resolve to existing pages
Three dashboard mode buttons	✓ PASS	Amateur / Starcom / ICS modes all present and switching
Starcom as standalone module	✓ PASS	Starcom cards in dedicated starcom-section; not in amateur-section
?mode=starcom URL parameter	✓ PASS	Dashboard auto-switches to Starcom mode on load
Planning P — 15 phases	✓ PASS	15 buttons, 5 groups, 100 objectives dropdown, generate sheet
ICS form POST to /api/forms	✓ PASS	nts.html, ics213.html, ics214.html all POST to correct endpoint
tactical.html localhost APRS ports	✓ PASS	localhost:8080/8082 for Graywolf/YAAC is intentional (Pi-local)
Dual UTC + local clocks on dashboard	✓ PASS	Both clocks tick every second in hero bar
Org name MCESV/MCEMA	✓ PASS	Updated across all 12 affected files and all 3 PDFs
apply_theme.py	✓ PASS	Scans/applies 14 CSS variables; all note types correct
ics_pdf_downloader.py	✓ PASS	22 FEMA ICS forms catalogued; --list/--status/--force work
install.sh syntax	✓ PASS	bash -n passes with zero errors; 21 installation steps
update.sh syntax	✓ PASS	bash -n passes; 12 menu options including new a/t options
fcc-refresh.timer	✓ PASS	Weekly refresh Sunday 03:00; Persistent=true
repeater-refresh.timer	✓ PASS	Monthly refresh 1st of month 04:00; token env file supported

SECTION 4 — AUDIT METHODOLOGY

METHOD	DESCRIPTION
API endpoint cross-reference	Every fetch() and XMLHttpRequest in all 30 HTML pages was matched against the server-side path handlers in all Python services. Mismatches flagged as bugs.
Host/port consistency check	All API base URLs scanned for localhost vs 192.168.50.1. Pages serving clients on EMCOMM-NET must use 192.168.50.1.
Database schema verification	All SQL table names used in Python servers cross-referenced against CREATE TABLE statements in db.py. Missing tables flagged.
Systemd ExecStart verification	Each ExecStart path in all 14 service units verified to point to an existing Python script or system binary.
HTML syntax check	All 30 HTML files parsed with Python HTMLParser. Unclosed tags and mismatched elements flagged.
Python syntax check	All 12 Python scripts compiled with python3 -m py_compile. Zero errors found.
Shell syntax check	install.sh and update.sh checked with bash -n. Zero errors found.
Inter-page link verification	All href= values in HTML pages checked against the file list. Template literals (\${f.link}) are excluded as they are runtime-generated.

Audit completed June 21, 2026. System is ready for beta testing on Pi 5 hardware. All critical and medium issues have been resolved. No open defects remain. The next step is physical hardware deployment and end-to-end testing on EMCOMM-NET.